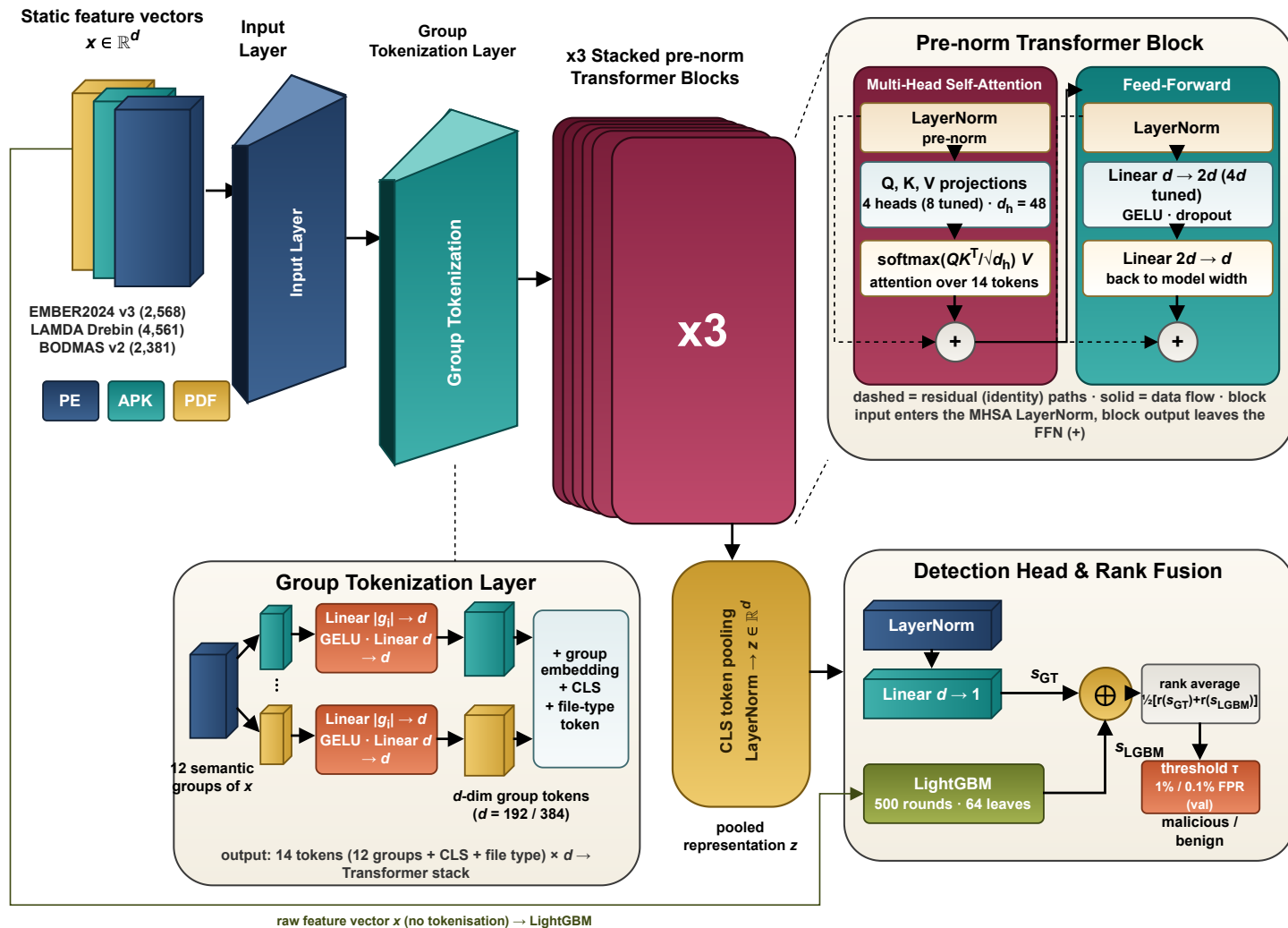




Benchmarks, Tuning & Evaluation Protocol

Benchmark Datasets

EMBER2024

3.2M files · 6 formats · 52 wk train / 12 wk test · challenge set 6,315

LAMDA

1.0M Android APKs · 2013-2025 · IID / NEAR / FAR

BODMAS

134K PE files · Aug 2019-Sep 2020 · monthly test

Optuna tuning of GT-Former

d , depth, heads, FFN, dropout, lr, wd, imports sub-tokens · objective: val ROC-AUC

5 random seeds per deep model
paired bootstrap (1,000 resamples)

Operational metrics

TPR @ 1% and 0.1% FPR · challenge-set detection · novel-family, singleton and low-consensus strata · weekly / monthly / yearly drift · confusion matrices, ROC, PR, MCC

Explainability

CLS attention over group tokens · SHAP on LightGBM · FR-MoE gate usage

Training

AdamW (lr 6e-4 tuned, one-cycle) · bf16 · single GPU · BCE loss